

1 The scientific method

EXPERIMENT 1

The scientific method

Goal

The goal of this experiment is to understand how the scientific method works. To do this you will make observations, look for patterns and come up with a hypothesis to *identify an unknown compound*. You will do this by comparing the reaction of the unknown compound with a set of known-reference-compounds, also called blanks. In this experiment, you will obtain some blank chemicals and an unknown. Your unknown is the same as one of the blanks. You will test each of these chemicals with a set of reagents. You have to observe the result of the test and record any change, such as *gas evolution*, the formation of a *precipitate*, a *change of color*, or maybe *no reaction*. Simultaneously, you will run similar tests for the unknown sample that is identical to one of the blanks. You will determine the unknown's identity by matching its characteristic reactions with one of the blanks.

Materials

- ☐ Six test tubes.
- ☐ The blanks: NaCl(s), NaI(s), Na₂CO₃(s), Na₂HPO₄(s) and Na₂SO₄(s).
- ☐ Unknown compound.
- ☐ The reagents: HNO₃(aq), Ba(NO₃)₂(aq), AgNO₃(aq) and thymol blue.

Background

Covalent compounds are chemicals resulting from the combination of nonmetallic elements. And example is CO₂, which results from combining carbon (a nonmetal) with oxygen (a nonmetal).

Properties of Covalent compounds

At normal conditions, covalent compounds may exist as solids, liquids, or gases. Covalent compounds do not exhibit any electrical conductivity, either in pure form or when dissolved in water. A typical covalent compound would be H₂O, water. Atoms in a covalent compound are connected by means of a covalent chemical bond. In a covalent bond, both atoms connected share the electrons. As an example, the HCl molecule has an hydrogen and a chlorine atom connected by means of a covalent bond, in which each atoms share the electrons of the bond.

Covalent naming

To name a covalent compound you need to (a) use the name of the first element in the compound, (b) use the first syllable of the second element, and (c) finish the name of the molecule in the suffix *-ide*. More importantly, you need to use prefixes (see Table 1) that indicate the number of atoms in the molecule. See the Table 1 below for a list of the different equivalencies between prefixes and numbers. As an example, the formula CH₄ is named carbon tetrahydride. Similarly, a covalent chemical name can be translated into a chemical formula (we call this to formulate a chemical with a given name), and the formula for carbon monoxide would be CO. Another example would be N₂O named as dinitrogen oxide, and the name sulfur hexafluoride corresponds to the formula SF₆. The prefix mono is omitted in the first element of the name, and for example, you will not name the chemical CO as ~~monocarbon~~ monoxide, you would just say carbon monoxide. When the vowels *a* and *o* appear together, the first vowel is omitted as in pentoxide instead of ~~penta~~oxide. A final example of a covalent compound would be:

Table ?? Prefixes used to name covalent compounds

Prefix	number	Prefix	number	Prefix	number
Mono	1	Tetra	4	Hepta	7
Di	2	Penta	5	Octa	8
Tri	3	Hexa	6	Nona	9

Example

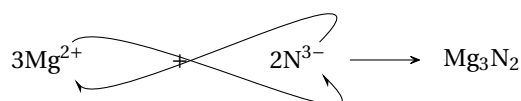
Name or formulate the following covalent chemicals: (a) NO (b) CS₂ (c) Sulfur Dioxide (d) Nitrogen Trichloride

Answer: All chemicals in this example are covalent as they result of the combination of nonmetals. In order to name them, we need to use prefixes and finish the suffix with -ide. The first chemical is called nitrogen monoxide. CS₂ is called carbon disulfide. The formula for sulfur dioxide and nitrogen trichloride are respectively SO₂ and NCl₃.

Ionic compounds result from the combination of a nonmetallic element with a metallic element. An example is NaCl, which results from combining sodium (a metal) with chloride (a nonmetal). Ionic compounds normally have high melting points and are solid under normal conditions. A typical ionic compound would be NaCl, cooking salt. Atoms of an ionic compound are connected through an ionic bond. In an ionic bond, one element gives away (the cation) and the other one receives electron density (the anion). As an example, in the NaCl molecule Na gives away electron density to Cl, and the molecule results from the combinations of Na⁺ and Cl⁻. In an ionic compound, the element on the left is positive and the one on the right is negative. Atoms of an ionic compound are connected through an ionic bond. In an ionic bond, one element gives away (the cation) and the other one receives electron density (the anion). As an example, in the NaCl molecule Na gives away electron density to Cl, and the molecule results from the combinations of Na⁺ and Cl⁻. In an ionic compound, the element on the left is positive and the one on the right is negative.

Combining ions

Ionic compounds are the result of combining two ions: a positive (cation) and a negative (anion) ion. Each ion has a charge, depending on its location on the periodic table. When combining two atoms you first need to arrange the ions starting from positive and followed by negative. The charges of an ion would become the coefficient of the other ion. For example Mg²⁺ and N³⁻ are combined as Mg₃N₂:



Another example would be the combination of Na⁺ and O²⁻ that would be Na₂O. You need to simplify the indexes of the formula by dividing by the smallest one, always using integer values. For example, Mg²⁺ and O²⁻ combine to give Mg₂O₂ that should be written as MgO



Another naming example that involves simplifying the formula is the chemical result of combining Ca²⁺ and C⁴⁻. After combining the charges we obtain Ca₄C₂ that needs to be simplified by dividing by the smallest number leading to Ca₂C.

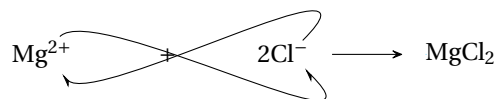
Simple ionic naming (type I ionic)

Type I ionic compounds result from the combination of a metal with given valence (Li, Ca, Mg, etc, see Figure ??) and a nonmetal. To name an ionic compound (type I ionic) you need to (a) use the name of the first element in the compound, (b) use the first syllable of the second element, and (c) finish the name of the molecule in the suffix -ide. As an example, the formula NaCl is named sodium chloride, and MgCl₂ is named magnesium chloride. Another example would be:

calcium chloride

CaCl₂ (ionic)

To formulate an ionic compound from a given name, we need to combine both ions by exchanging the valences (the ionic charges). For example, MgCl₂ results from the combination of Mg²⁺ and Cl⁻ so that the number 2 in MgCl₂ near the Cl atom is coming from the Mg²⁺. In other words:



The sign of the charges only indicates which element goes first in the formula: the positive element (cation) first followed by the negative element (anion). For example, the result of combining Na⁺ and Cl⁻ is NaCl and not ClNa as Na has a positive ionic charge and has to appear first in the formula.

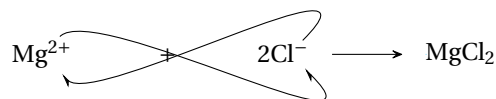
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Complex ionic chemicals

NaCl results from the combination of Na⁺ and Cl⁻. The ionic charges of Na and Cl are given in Figure ?? according to the group. If the ionic chemical contains a transition metal with variable ionic charge, that is, which is not in Figure ?? then the ionic naming becomes a bit more complex. The reason is that one needs to specify the charge of the metal, explicitly in the name of the chemical. An example would be NiCl₂ named as Nickel(II) chloride or Co₂O₃ named as Cobalt(III) oxide.

Name complex ionic chemicals

This section covers how to name ionic chemicals containing a metal with variable charge. In this case, you need to specify the charge of the metal in the name. To calculate this number you will solve a simple math equation. For example, the name of Mn₂O₃ is Manganese(III) oxide. How do we get this name? Manganese has several charges as it is not in Figure ??, let us use x for its charge Mn^x and oxygen has a charge of two O²⁻. After combining Mn^x and O²⁻ the resulting formula would be Mn₂O_x. By comparison with the given formula, Mn₂O₃, x has to be three and hence the charge of Mn has to be three. Therefore, the final name would be Manganese(III) oxide.

Example

Name or formulate the following ionic compounds: (a) MnO (b) Fe₃N₂ (c) Cobalt(II) carbide (d) Iron(II) oxide

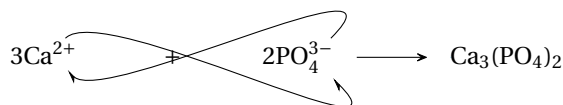
Answer: All the chemicals on this example contain a metal that can have several charges, and hence, we need to specify the ionic charge on the name. MnO results from Mn^x and O²⁻. After combining the ions, the formula would be Mn₂O_x, a formula that needs to be compared to MnO. The formulas do not look similar, so let's make them more similar

by dividing by two so that MnO_x resembles MnO . By comparing x has to be 2 and hence the name is Manganese(II) oxide. The name for Fe_3N_2 would be Iron(II) nitride. The valence of Iron comes from combining Fe^x and N^{3-} that gives Fe_3N_x . By comparison with Fe_3N_2 x has to be two and the name is Iron(II) nitride. the formula for Cobalt(II) carbide would be Co_2C as Cobalt(II) is Co^{2+} and carbide is C^{4-} . After combining the ions one obtains Co_4C_2 that gives Co_2C . Finally, the formula for Iron(II) oxide is FeO as Iron(II) is Fe^{2+} and oxide is O^{2-} that gives Fe_2O_2 and simplifying one obtains FeO .

At this point, we saw the naming and formulation of ionic (e.g. NaCl) and covalent compounds (e.g. CO_2). This section covers the naming of complex salts: oxosalts and hydrosalts. In general, salts (oxosalts or hydrosalts) are the result of mixing an oxoacid and a base. Their formulas tend to look more complex than simple ionic or covalent compounds as they are made of at least three different elements. An example of an oxosalt would be CaSO_4 called calcium carbonate. An example of hydrosalt would be NaHSO_4 which is called sodium monohydrosulfate. This section will also cover the naming of hydrates (e.g. $\text{CaSO}_4 \cdot \text{H}_2\text{O}$), which are compounds containing water molecules inside their structure. Before being able to name these complex chemicals it is convenient to practice combining ions, without paying attention to the chemical naming.

Combining ions

To combine two ions, you first arrange the positive ion on the left followed by the negative ion on the right, to then cross the ionic charges from the top of the ion to the bottom of the opposite ion. The positive and negative charges are not carried. If the ions have more than one element we have to use parenthesis. An example would be combining Ca^{2+} and PO_4^{3-} leading to $\text{Ca}_3(\text{PO}_4)_2$:



We would need to simplify the formula in case the charges compensate each other.

An example would be combining Mg^{2+} and SO_4^{2-} leading to MgSO_4



Naming Oxosalts

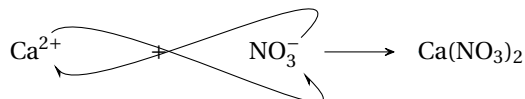
The names of the oxosalts are constructed by combining the name of the first element—you need to specify its charge in the case of a transition metal element with different possible charges—followed by the name of the oxosalt from Table 2. For example, the name of MgSO_4 is magnesium sulfate, as Mg^{2+} is magnesium and SO_4^{2-} is sulfate. Another example is $\text{Fe}_2(\text{CO}_3)_3$ called Iron(III) carbonate. A final example:

Litium nitrate

LiNO_3 (oxosalt)

Formulating Oxosalts

When you know the name of an oxosalt and you want to obtain its formula, you first need to arrange the positive ion on the left followed by the negative ion on the right, to then cross the ionic charges from the top of the ion to the bottom of the opposite ion. For example, calcium nitrate results from the combination of Ca^{2+} calcium and NO_3^- , nitrate. By combining the two ions we obtain the final formula as $\text{Ca}(\text{NO}_3)_2$:



Naming oxosalts with hydrogen

Some oxosalts contain hydrogen atoms in their structure between the oxosalt cation and anion (e.g. NaHSO_4). For example, NaHSO_4 is named sodium monohydrogensulfate. To understand this name, we will first focus on the second part of the name, monohydrogensulfate which represents the anion. The name monohydrogensulfate (HSO_4^-) comes from adding a proton (H^+) to a sulfate cation (SO_4^{2-}). Mind that protons (H^+) are positively charged and therefore if we add a single H^+ to a sulfate cation (SO_4^{2-}) the charge will have to decrease a single unit, giving us HSO_4^- . As we can see, the name is directly related to the oxosalt anion and the number of hydrogens in the hydrosalt name. For example, phosphate (PO_4^{3-}) is an oxosalt anion whereas hydrogenphosphate (HPO_4^{2-}) and dihydrogenphosphate (H_2PO_4^-) are both hydrosalt anions. An explanation about the charges: as phosphate has three negative charges, hydrogenphosphate has to have one less charge (that is 2-) and dihydrogenphosphate has to have two less negative charges (that is -1). Some final hydrosalt anions examples:

carbonate
monohydrogen carbonate

CO_3^{2-} (oxosalt ion)
 HCO_3^- (hydrosalt ion)

Above we saw how to name just the ending of the oxosalt with hydrogen anion. Now we can move forward to the whole naming of the salt. We just need to add the name of the element in the first place, and for example, NaH_2BO_3 would be named sodium dihydrogenborate. If the first ion—the cation—is a transition metal cation (a type two cation) we need to include in parenthesis the valence of the cation. For example, $\text{Fe}(\text{H}_2\text{BO}_3)_2$ would be named iron(II) dihydrogenborate. More examples:

sodium carbonate
sodium monohydrogen carbonate

Na_2CO_3 (oxosalt)
 NaHCO_3 (hydrosalt)

Hydrates

Some chemicals contain water molecules trapped in their structure and therefore water molecules (H_2O) are often indicated in chemical formulas. These types of chemicals containing water are called *hydrates*, precisely because hydrate means water. Examples of hydrates are $\text{BeSO}_4 \cdot 4\text{H}_2\text{O}$ or $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ called respectively beryllium sulfate tetrahydrate and copper(II) sulfate pentahydrate. To formulate hydrates you just need to use prefixes such as mono, di, tetra—the same ones we use to name covalent chemicals, see Table 1—to indicate the number of water molecules in the chemical and end the name with *hydrate*. As a note, warming up hydrates (e.g. $\text{BeSO}_4 \cdot 4\text{H}_2\text{O}$) results in the release of water producing a dehydrated or *anhydrous* compound (e.g. BeSO_4). A final example of hydrate naming:

Sodium sulfate pentahydrate

$\text{Na}_2\text{SO}_4 \cdot 5\text{H}_2\text{O}$ (a hydrate)

Example

Name or formulate the following oxosalts: (a) K_2SO_4 (b) Na_2CO_3 (c) Magnesium carbonate (d) Sodium phosphate

Answer: K_2SO_4 is named potassium sulfate, as K^+ is potassium and SO_4^{2-} stands for sulfate. Na_2CO_3 is sodium carbonate. Magnesium carbonate is MgCO_3 and sodium phosphate is Na_3PO_4 .

Table 2 Names of oxoacids and oxoalts (top table) and hydracids (bottom table).*				
Element	Oxoacid	Oxoacid Name	Oxoasalt	Oxoasalt Name
Manganese	HMnO ₄	Permanganic Acid	MnO ₄ ⁻	Permanganate
	H ₂ MnO ₄	Manganic acid	MnO ₄ ⁻²	Manganate
Carbon	H ₂ CO ₃	Carbonic Acid	CO ₃ ⁻²	Carbonate
Nitrogen	HNO ₃	Nitric Acid	NO ₃ ⁻	Nitrate
	HNO ₂	Nitrous Acid	NO ₂ ⁻	Nitrite
Phosphorus	H ₃ PO ₄	Phosphoric Acid	PO ₄ ⁻³	Phosphate
Sulfur	H ₂ SO ₄	Sulfuric Acid	SO ₄ ⁻²	Sulfate
	H ₂ SO ₃	Sulfurous Acid	SO ₃ ⁻²	Sulfite
	H ₂ S ₂ O ₂	Thiosulfurous Acid	S ₂ O ₂ ⁻²	Thiosulfite
	H ₂ S ₂ O ₃	Thiosulfuric Acid	S ₂ O ₃ ⁻²	Thiosulfate
	H ₂ S ₂ O ₇	Disulfuric acid	S ₂ O ₇ ⁻²	Disulfate
Chlorine	H ₂ S ₂ O ₈	Peroxodisulfuric acid	S ₂ O ₈ ⁻²	Peroxodisulfate
	HClO ₄	Perchloric Acid	ClO ₄ ⁻	Perchlorate
	HClO ₃	Chloric acid	ClO ₃ ⁻	Chlorate
	HClO ₂	Chlorous acid	ClO ₂ ⁻	Chlorite
Iodine	HClO	Hypochlorous acid	ClO ⁻	Hypochlorite
	HIO ₄	Periodic Acid	IO ₄ ⁻	Periodate
Chromium	H ₂ CrO ₄	Chromic acid	CrO ₄ ⁻²	Chromate
	H ₂ Cr ₂ O ₇	Dichromic acid	Cr ₂ O ₇ ⁻²	Dichromate
Boron	H ₃ BO ₃	Boric acid	BO ₃ ⁻³	Borate
Hydracid	Hydracid Name	Hydracid	Hydracid Name	
HCl	Hydrochloric acid	HBr	Hydrobromic acid	
HI	Hydroiodic acid	HF	Hydrofluoric acid	
HCN	Hydrocyanic acid	H ₂ S	Hydrosulfuric acid	

* Yellow indicate very important acids

Procedure

Part A: Indicate your hypothesis

- ☐ *Step 1:* – Obtain the unknown compound, the blanks, and the reagents.
- ☐ *Step 2:* – Compare the unknown with the blanks and guess which blank looks like the unknown. That is your hypothesis. Write down the name of your guess on the results page.
- ☐ *Step 3:* – Wash a 100 mL graduated cylinder and 400 mL beaker. Place a pea-size portion of the unknown in the beaker, and add 150 mL of water using the cylinder to measure it. Swirl until all the solid has been totally or partially dissolved.
- ☐ *Step 4:* – Set aside the remaining portion of your unknown.

Good Lab Practice

Obtain directions from your lab instructor for discharging the solutions that you will use in this experiment.

Part B: Testing for gas

- ☐ *Step 1:* – In this first experiment you will use the solid version of the blanks, not the droppers. Obtain six test tubes. In each test tube, you will add a blank reagent in a solid state, that is, one blank per test tube. To each test tube, you will add a pea-size sample of a different blank in a solid state using its spatula. You will add the unknown in the sixth test tube. This way, place a pea-size solid sample of NaCl in the first test tube. Add a pea-size sample of NaI

in the second test tube, and so on with the rest of the blanks, Na_2CO_3 , Na_2HPO_4 , Na_2SO_4 and your solid unknown in the following test tubes. Mark your test tubes for recognition.

- ☐ *Step 2:* – Add five drops of HNO_3 ⚠ to each test tube and record your observations in the results table: *GE*, for gas evolution, *NGE* for no gas evolution.
- ☐ *Step 3:* – Discharge the solutions from the test tubes into the right waste container. Wash the test tubes, and rinse them with distilled water.

Part C: Testing for a barium precipitate

- ☐ *Step 1:* – For the following test will use only liquid samples in the dropper, not the powders. Use the solutions of NaCl , NaI , Na_2CO_3 , Na_2HPO_4 and Na_2SO_4 found in the laboratory, and the solution of your unknown prepared in Step 3 for the remaining tests in this practice.
- ☐ *Step 2:* – Using six clean test tubes, add 20 drops of each of the solutions as well as the unknown to separate test tube. In short, five of the test tubes should have a different blank and the sixth test tube should have the unknown solution. Add 3 drops of a solution of ammonia (NH_3) in each test tube ⚠ and swirl until solved.
- ☐ *Step 3:* – Add five drops of the $\text{Ba}(\text{NO}_3)_2$ ⚠ solution to each of the test tubes. Shake gently to obtain homogeneity. Examine each test tube carefully and look at the results. Record your observations: *P* for precipitate, *NP* for no precipitate.

⚠ CAUTION!

⚠ Nitric acid (HNO_3) and ammonia (NH_3) ⚠ can cause chemical burns.

⚠ Solutions containing barium are toxic.

Part D: Testing for silver precipitate

- ☐ *Step 1:* – Using six clean test tubes, add 20 drops of each of the solutions including the unknown to separate test tubes (e.g. NaCl , NaI , Na_2CO_3 , Na_2HPO_4 , Na_2SO_4 , and unknown). In short, five of the test tubes should have a different blank and the sixth test tube should have the unknown solution.
- ☐ *Step 2:* – Add five drops of the nitric acid AgNO_3 solution to each test tube. Record your observations: *P* for precipitate, *NP* for no precipitate. Include the color of the precipitate.

Part E: Testing with thymol blue

- ☐ *Step 1:* – Using six clean test tubes, add 20 drops of each of the solutions including the unknown to separate test tubes (e.g. NaCl , NaI , Na_2CO_3 , Na_2HPO_4 , Na_2SO_4 , and unknown). In short, five of the test tubes should have a different blank and the sixth test tube should have the unknown solution.
- ☐ *Step 2:* – Add five drops of the thymol blue solution to each test tube. Shake each test tube gently before recording your observations: *write down Blue, Yellow green, or Red.*

Part F: Identifying the unknown compound

- ☐ *Step 1:* – The unknown is one of these compounds: NaCl , NaI , Na_2CO_3 , Na_2HPO_4 , Na_2SO_4 . You should be able to identify your unknown compound by matching its reactions with those of one of the known samples.
- ☐ *Step 2:* – When done with the experiment, clean up all test tubes using a test tube brush and soapy water.

STUDENT INFO

Name: _____ Date: _____

Pre-lab Questions**The scientific method**

1. Classify the following compounds as *ionic, covalent, acids, bases* or *oxo salts*:

NH ₃	_____	SO ₃	_____
HNO ₃	_____	NaF	_____

2. Give the names or formulas for the following compounds:

NaCl	_____	HNO ₃	_____
Sodium iodide	_____	Sodium carbonate	_____

3. Name or give the formula of the following compounds:

SO ₂	_____	NO ₂	_____
Lithium fluoride	_____	Sodium oxide	_____

4. What does a pea size of solid mean?

5. Where in the lab you will be disposing of the disposals from this experiment?

6. The following table resulted from an experiment, very similar to the one you are going to carry. The goal was to identify an unknown compound between KCl and KI. Look at the reactivity patterns and identify the unknown.

		Reagents			
		HNO ₃	Ba(NO ₃) ₂	AgNO ₃	Thymol Blue
Blanks	KCl	NP	NP	P _(white)	Green
	KI	NP	NP	NP	Green
	Unknown	NP	NP	NP	Green

My unknown is = _____

STUDENT INFO

Name: _____ Date: _____

**Results
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Unknown Label # = _____ Your guess = _____

		Reagents			
		HNO ₃	Ba(NO ₃) ₂	AgNO ₃	Thymol Blue
Blanks	NaCl				
	NaI				
	Na ₂ CO ₃				
	Na ₂ HPO ₄				
	Na ₂ SO ₄				
	Unknown				

(write GE for gas evolution, NGE for no gas evolution, P for precipitate, NP for no precipitate and Blue, Yellow or Green for the color for the Thymol blue experiment)

My unknown is = _____

STUDENT INFO

Name: _____ Date: _____

Post-lab Questions

The scientific method

1. Classify the following compounds as *ionic*, *covalent*, *acids*, *bases* or *oxo salts*:

Ba(OH) ₂	_____	CO ₂	_____
HCl	_____	H ₂ SO ₄	_____
LiCl	_____	AuCl	_____
Na ₂ SO ₄	_____	NaOH	_____

2. Give the names or formulas for the following compounds:

HNO ₃	_____	Sodium sulfate	_____
Barium nitrate	_____	Na ₂ CO ₃	_____
AgNO ₃	_____	Nitric acid	_____

3. Name or give the formula of the following compounds:

CuCl ₂	_____	Hydrogen chloride	_____
Nickel (II) chloride	_____	BaO	_____
CoO	_____	MgO	_____
Sodium iodide	_____	Potassium iodide	_____
NH ₃	_____	LiH	_____

